

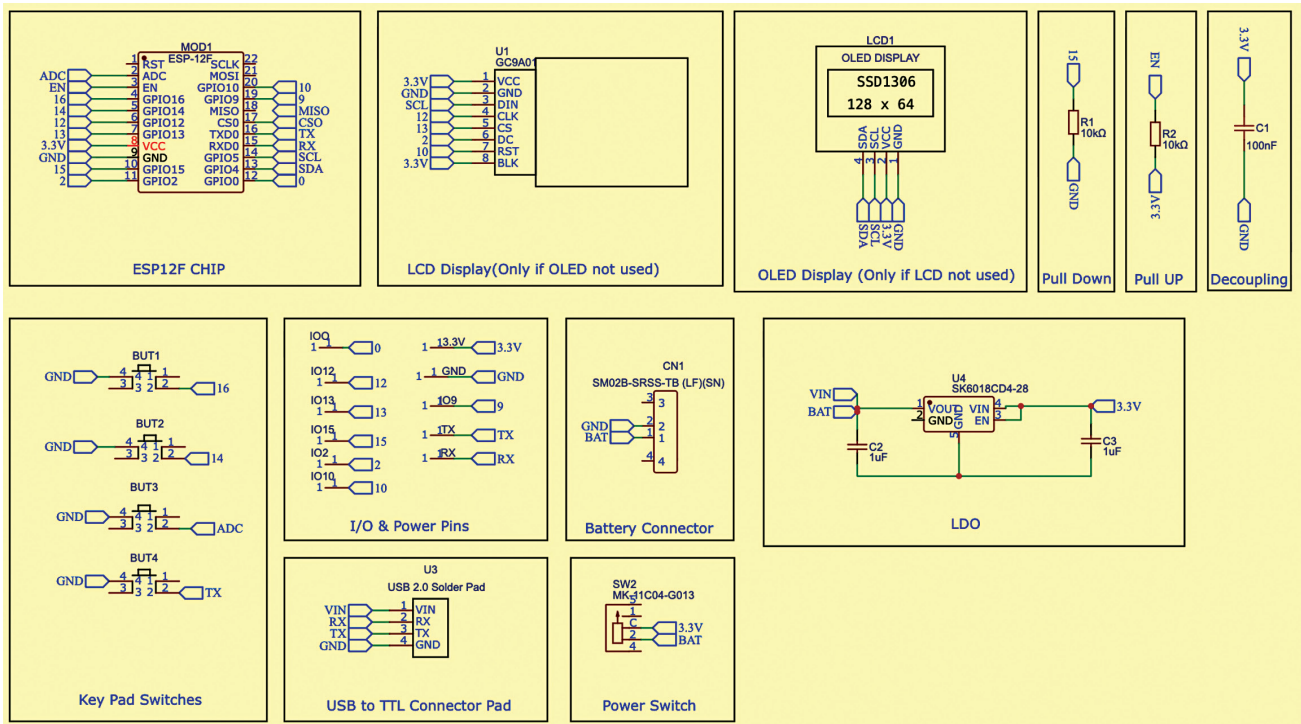


Fig. 6: Complete circuit

GPIO0 must be pulled low; a jumper is provided to force this pin low only during upload, while it remains pulled high during normal operation. The EN (CH_PD) pin is pulled up to VCC via a resistor to keep the chip enabled.

For stable booting

To ensure stable booting, follow these steps:

1. Pull GPIO15 down to GND for normal boot using R1 (10kΩ).
2. Pull GPIO0 low only during programming to enable firmware upload; in normal operation, it remains high.
3. Keep GPIO2 pulled up to ensure the correct boot configuration.
4. Pull EN (CH_PD) up to 3.3V using R2 (10kΩ) to keep the chip enabled.

A decoupling capacitor, C1 (100nF), should be placed close to VCC and GND to filter high-frequency noise and stabilise the power rail during Wi-Fi transmission bursts. The module features dedicated VCC (3.3V) and GND pins. With appropriate pull-up,

pull-down, and decoupling networks, the ESP12F operates reliably as the core of the portable device.

Peripheral connections use the remaining GPIOs and are routed for either an OLED display (I²C: SDA, SCL) or SPI colour displays (MOSI, SCLK, CS, DC) along with button inputs for user control. Fig. 4 shows the ESP12F footprint and pinout, and Fig. 5 shows the chip configuration.

Complete circuit

Fig. 6 shows the complete circuit diagram of the portable Game Boy and Wi-Fi hacking device. For clarity, the circuit is divided into subsections.

Keypad buttons. Four tactile push-buttons provide game controls, menu selection, and basic UI navigation. Buttons are mapped to available GPIO pins; the design is expandable, for instance, via an I²C-based touch keypad. Refer to Fig. 7 for the complete circuit.

Battery connector. A SM02B-SRSS-TB connector accepts a single-cell Li-ion or LiPo battery. The connector routes BAT and GND to the

LDO regulator, providing a stable 3.3V supply to the ESP12F and the display, supporting a compact, wearable form factor.

USB to TTL programming pad.

A USB 2.0 solder pad exposes VIN, RX, TX, and GND for connection to an external USB-to-TTL adaptor (FTDI/CP2102/CH340), enabling firmware flashing and debugging without additional hardware.

Power switch. A slide switch (SW2-MK-51C04-G013) is placed between the 3.3V regulator output and the battery input to disconnect the battery when not in use, thereby preventing unnecessary drain.

I/O and power pins. Unused GPIOs (IO0, IO2, IO9, IO10, IO12, IO13, IO15, TX, RX) plus 3.3V and GND are routed to a header for expansion with external modules, sensors, or additional controls. This extends functionality beyond gaming to prototyping and IoT applications.

PCB design

The ESP12F module is placed on the bottom layer with the USB